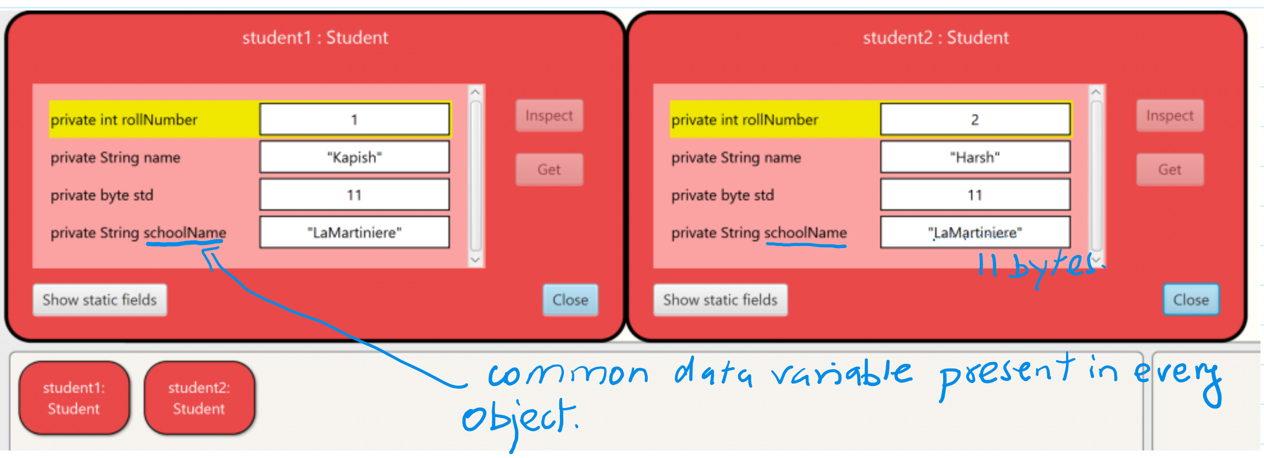
"static" keyword in Java

08 May 2026 05:47

What is the static keyword in Java?
The static keyword in Java is used for memory management mainly. We can apply static keyword with variables, methods, blocks and nested classes. The static keyword belongs to a class then an instance of a class.

```java
package OOP;


/**
 * Write a description of class Student here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class Student implements Cloneable
{
    private int rollNumber;
    private String name;
    private byte std;
    //Constructor Chaining
    public Student(){ //default constructor defined by the user
        this(0,"",(byte)0); //calling parametrized constructor
    }

    //parametrized constructor, used to initialize instance variable of object
    public Student(int rn, String nm, byte std){
        this.rollNumber = rn;
        this.name = nm;
        this.std = std;
    }

    public Student(Student st){   //copy constructor
        this(st.rollNumber, st.name, st.std); //calling parametrized constructor
    }

    public void setRollNumber(int rollNumber){
        this.rollNumber = rollNumber;
    }

    public void setName(String nm){
        this.name = nm;
    }

    public void setStd(byte std){
        this.std = std;
    }

    public int getRollNumber(){
        return this.rollNumber;
    }
```

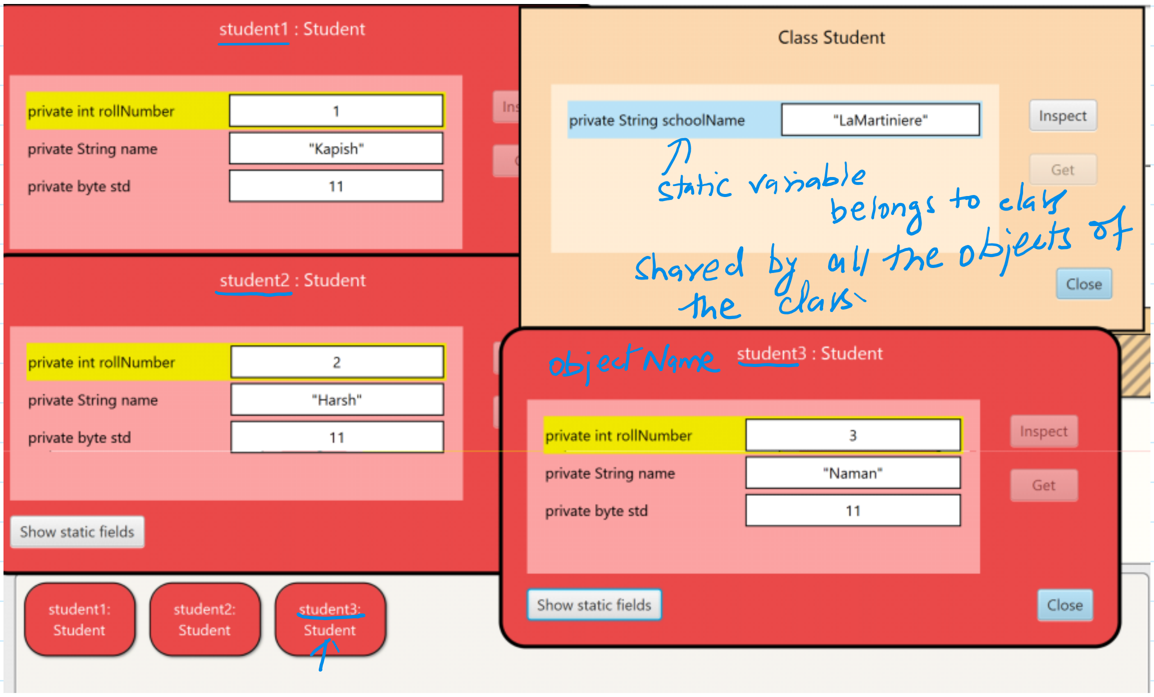
```
    public String getName(){
        return this.name;
    }

    public byte getStd(){
        return this.std;
    }

     // Overriding the clone() method
    @Override
    public Object clone() throws CloneNotSupportedException {
        // Returning a clone of the current object
        return super.clone();
    }

    //factory method
    public static Student newStudent(){
        return new Student();
    }

    @Override
    public String toString(){
        return "\nRoll number: " + rollNumber +
                "\nName " + name + "\nStd: " + std;
    }
}
```



11 bytes.

common data variable present in every object.

5000 x 11 = 55000 bytes.

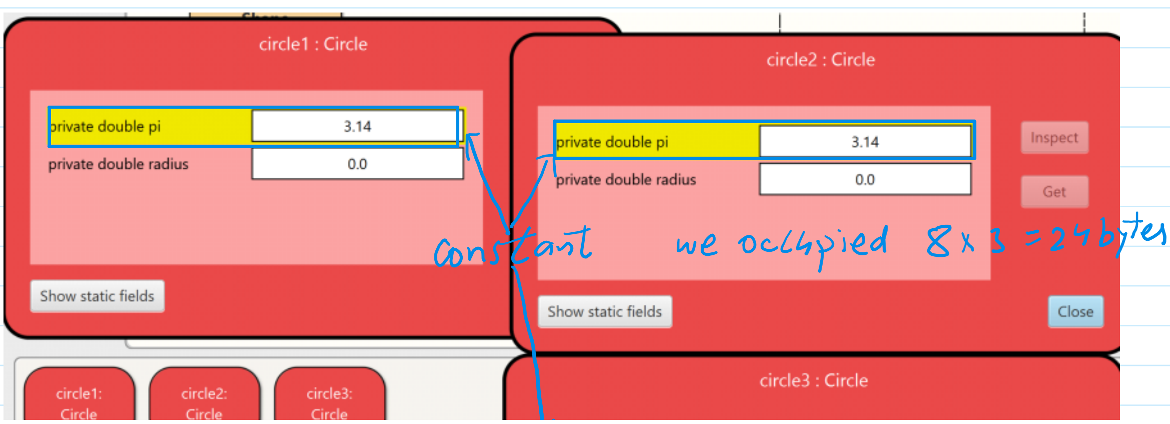


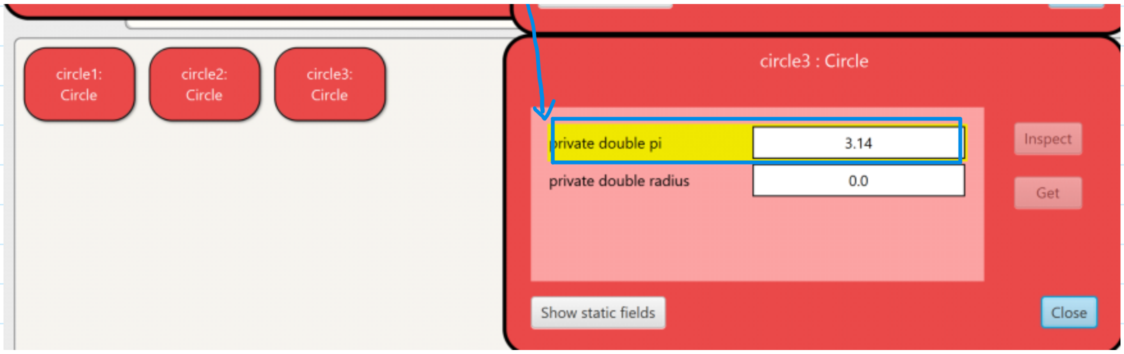
Types of Static Members
The static keyword can be used with the following:
1. Variable(also known as a class variable)
2. Method (also known as class method)
3. Block
4. Nested class

Uses of static keyword
Beyond memory management, the static keyword in Java has several other users. When it comes to variable, it means that all instances of a class share the variable and belongs to the class as a whole, not just anyone instance.

Static methods are available throughout the program since they can be called without first generating an instance of a class. When the class loads, static blocks are used to initialize static variables or carry out one time operations.

Furthermore, nested static classes can be instantiated individually, but are still linked to the outer class. Moreover, class names can be used to access static variables and methods directly, eliminating the need to build an object instance. This is especially helpful for constants or utility methods.





Characteristics of static Keyword

1. **Belongs to the class, not the object:** Anything marked as static is tied to the class itself, not individual objects. It means all objects share the same static variables and methods.

2. **Access without an object:** Since static members belongs to the class, you can access them directly using the class name. (Class_name.methodName()) No need to create an object.

3. **Static methods cannot use instance variables directly:** A static method does not know about specific objects, so it cannot directly access non-static (instance) variables or methods.

4. **Main method is static:** The main method in Java is static so that Java can run the program without creating an object first.

5. **Executed once when the class is loaded:** A static block runs once when the class is first loaded, making it useful for initializing things like constants.

6. **Not Overridden like normal methods:** Static method do not follow typical method over writing. If a subclass defines the same static method, it's more like redeclaring it rather than overriding it.

Java Static Variable

If we declare any variable as static, it is known as a static variable.

- The static variable can be used to refer to a common property of all objects (which is not unique for each object.) For example, school name of students Or company name for employees.
- The static variable gets memory only once in the class area at the time of class loading.
- Static variables in Java are also initialized to default values if not explicitly initialized by the programmer. They can be accessed directly using the class name without needing to create an instance of the class.
- Static variables are shared among all instances of the class. Meaning if the value of static variable is changed in one instance, it will reflect the change in all other instances as well.

Java Static Method

If we apply a static keyword with any method, it is known as a static method.

- A static method belongs to a class rather than the object of a class.
- A static method can be invoked without the need for creating an instance of a class.
- A static method can access static data members and can change their value of it.

```java
public class Student implements Cloneable
{
    private int rollNumber;
    private String name;
    private byte std;
    private static String schoolName;

    public static void setSchoolName(String sn){
        schoolName = sn;
        this.std = 12;
    }
    //Co
    public Student(){ //default constructor defined by the user
        this(0,"",(byte)0); //calling parametrized constructor
    }
```

non-static variable this cannot be referenced from a static context

instance variable

class variable

method belongs to class (static method)

Restrictions for the Static Method

1. The static method cannot use non-static data members or call a non-static method directly.
2. 'this' and 'super' keyword cannot be used in static context.

Java Static Block

- It is used to initialize the static data member.
- It is executed before the main() method at the time of class loading.

```java
public class Student implements Cloneable
{
    private int rollNumber;
    private String name;
    private byte std;
    private static String schoolName;
```

```java
private int rollNumber;
private String name;
private byte std;
private static String schoolName;
static{
    System.out.print("\fSet your school name");
}
```

This block will execute only once, when we are going to create 1st instance of the class.